

The Inclusive Metaverse: Using Self-discrepancy Theory and Artificial Intelligence to Create Personalized Immersive Experiences

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Abstract

I am currently a second-year PhD student at the Nottingham School of Art and Design, Nottingham Trent University. My PhD thesis aims to investigate the relationship between artificial intelligence and user electroencephalography (EEG) data through studies exploring user needs in hybrid interactive experiences. To expand the potential connection between user experience design and neuroscience, I aim to design real-time interactive experiences that use EEG data as input. These AI-generated experiences both avoid brain fog, a concern among neuroscientists regarding frequent AI use, and they also train user concentration and thinking skills. I examine user experience as a fundamental principle that shapes interactive design using innovative input data formats. Self-discrepancy theory serves as the foundation for user experience in my thesis, letting users easily identify their unique and ideal experiences.

CCS Concepts

• Human-centered computing; • Human computer interaction (HCI); • Empirical studies in HCI;

Keywords

User experience design, Metaverse, Electroencephalography data, Generative artificial intelligence, Cozy game, Real time experience

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1 Background and motivation

Self-discrepancy theory [12] provides the foundation for the rationale of developing user experience in my thesis. Self-discrepancy theory is the leading theory that distinguishes between the ideal self that consumers expect and the actual and ought selves that they often express and are forced to express in real life. The degree

of correlation in this distinction further clarifies the potential benefits the Metaverse can offer for virtual interactive experiences, allowing users to freely interact and immerse themselves in an ideal virtual life that aligns with their expectations. Every human being has a desire to express or even live up to their ideal selves, especially the creative practitioners I am targeting as potential subjects of this research. Moreover, discovering the self-creation is a core motivating factor for people to express their innermost selves in their creativity.

User experience design is an interdisciplinary application of HCI and psychology [13]. However, developers typically base their experiences on a specific user group, and currently, they have not yet created personalized experiences tailored to each individual [7]. In the age of AI, it is no longer uncommon for users to receive personalized feedback; it happens in conversations where humans interact with ChatGPT or AI robots that personalize each user's device. All that personalization comes from developer training, and users make specific text, voice, or settings of requests.

AI has become an important part of many fields of life studies in general and HCI in particular. I have witnessed three major waves of HCI, including (1) viewing humans as part of a machine system, (2) viewing humans as cognitive agents, and (3) viewing humans as subjects playing a central role in interactive activities [10]. The fourth wave of HCI is a cross-disciplinary model [11], and it is predicted to serve as a guiding framework for interactive design values, as a high-level concern regarding values and ethics in human-machine interaction [15]. I aim to explore the positive values of AI in advancing the fourth wave of HCI in user experience design, shaping the experience toward important visions that serve society [5]. I research the potential for cognitive and creative abilities, as well as how people bring their intuitive emotions and personal social contexts into the experience.

Biofeedback is a process that allows an individual to make changes to human physiological activity to improve health and performance (Edmonds and Tenenbaum, 2011). It typically includes the use of sensors to monitor bodily functions such as heart rate, breathing rate, muscle tension (electromyography), galvanic skin response, skin temperature, and even brain activity (electroencephalography – EEG) [5], [3]. These biological signals are now being applied to generate real-time personalized responses with the involvement of artificial intelligence [3]. That result is also consistent with previous studies on how experiences usually include images and sounds that support users to control their physiological state proactively [6], [8]. Moreover, by understanding real-time user physiological



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data, researchers can use it to investigate and make findings about experiential interventions, adjusting for real-time data to provide users with an optimally personalized experience [3]. The EEG data in my thesis is not only personalized but also shows the mental state of people, which is highly applicable to understanding how the brain works or can be used to train the system and create images and sounds. More broadly, the way EEG data works can apply to user experience design in the form of a constantly changing state of mind that will impact the real-time feedback generated by *generative artificial intelligence* (GenAI). These orientations aim to design an “invisible” interaction model where humans and computers act as co-creators, merge into a flow of thought, reduce barriers between humans and computers, and enhance natural connectivity.

My thesis aims to show how applying the traditional process of user experience design expands the scope of input data from tangible to intangible forms by combining and exploring the applicability of innovative elements in creating user experiences. In addition, my thesis offers a *closed-loop immersive model* that can be expanded to various forms of experiential design. I will investigate the potential for developing immersive Metaverse experiences that allow users to participate in virtual environments where their ideal selves can be more easily explored and fulfilled than in their real lives. This research aims to design a metaverse model with essential framework elements to create experiences that help users mitigate the negative impacts of AI and social media brain fog in the digital world [1]. This research contributes to a more comprehensive metaverse, overcoming the negative aspects of the current digital world and paving the way for hybrid interactions.

2 Research objectives

Unlike current social media experiences or AI interactions that negatively affect the human brain, my research aims to design experiences that train users’ concentration and memory skills. These rich AI-generated experiences have the potential to inspire creativity and artistry in creative practitioners. I aim to demonstrate that AI can also have a positive impact on users’ brains when I design a highly valuable experience.

To provide valid arguments for the research, I pursued the following objectives:

- Delve into the experience designs built from *cozy games* to understand the UX and UI of a hybrid experience that can provide players with positive emotions and immersion [17].
- Target creative practitioners in the early stages of their careers, who are highly creative but not yet fully understand how creativity can benefit from technologies such as AI, Metaverse, and EEG.
- Target creative practitioners with over 20 years of experience in the arts to understand how they apply technology to enhance their creative performance; to investigate how professionals approach AI and integrate it into their products with critical thinking.
- Focus on user experience design to provide a connected experience between different users in real time, with EEG as input data.
- Design an immersive experience in a metaverse setting that incorporates innovative elements. The program captures

user focus through eye contact and uses Google VEO3 to generate real-time responses based on the areas the user is focusing on.

3 Methodology

My PhD thesis aims to examine both users’ current technology interaction experiences and their expectations for future experience features, to create a highly applicable design that leverages the positive value of AI and personalized user data.

I used a systematic review to assess papers on the research context, which was crucial for identifying gaps in current research that need to be filled. The term “netnography” refers to a research method that involves ethnographic study of technology and the internet. Netnography was introduced through online interaction, shaping the contemporary world [2]. I used thematic analysis to analyze six cozy games (with seven females and four males) and specific cozy game videos recorded and posted on YouTube by players: Bokura [4], Mail Time [14], Potion Craft [18], The Sims [19], Tiny Glade 1 [20], Tiny Glade 2 [21].

To gather diverse expectations from participants about the future user experience in the Metaverse created using GenAI and EEG data, I used semi-structured interviews and co-design as a method to collect information to facilitate the creation of typical scenarios, personas, and user flows, and to serve Study 2 in investigating the ethics of GenAI within the framework of developing an interactive Metaverse environment. I used thematic analysis, sharpening, sorting, focusing, and discarding methods [16] to analyze results from 29 participants, as well as patterned coding to categorize the unique characteristics and experiential needs of everyone.

In study 3, I conducted in-depth interviews with four experts (two females, two males) from different areas of the creative and artistic fields. I aimed to investigate their perspectives on the use of AI in artistic creation and how they use it daily in their work. Contextual thematic analysis was used to analyze each expert interview, understanding their individual contexts, and how they use AI. This study aims to give participants a direct perspective on their ideal self and create a specific mood board of the artist’s ideal self.

In study 4, I used observation to assess changes in participants’ facial expressions. I recorded participants’ electroencephalogram (EEG) data in real time, cleaned the data, and calculated EEG results for comparison with the observation results. I did post-study qualitative questions section to investigate participants’ thoughts on their experience and their expectations for improving the experience in the future. Linking qualitative and quantitative data was used to first compare results from participants’ experiences. When analyzing participants’ data, I focused on both analyzing the data and providing an overview of this study.

In Study 5, I will need to address the themes: (1) combining eye tracking to identify the area the user is focusing on and (2) training AI so that the system understands real-time EEG data. It is necessary to build a strong baseline for the experiment based on the accuracy of: (1) EEG data results, (2) post-study questionnaires, and (3) observing the user’s facial expression.



Figure 1: Thematic Coding for Netnography Data in Cozy Games

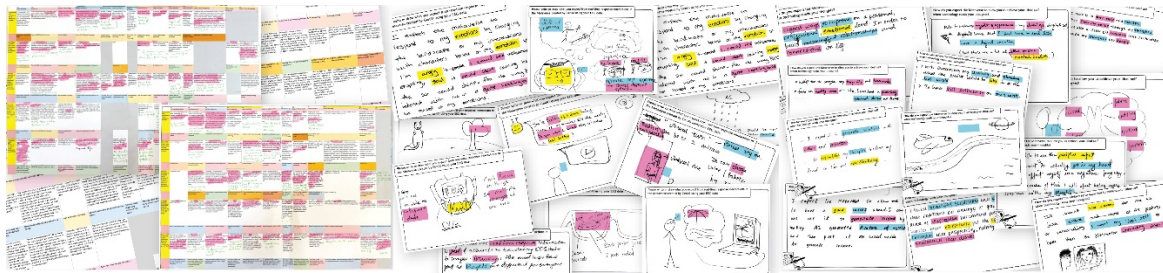


Figure 2: Data collected from participants in the co-design workshop

4 Results and contributions to date

During my first year of the PhD program, I focused on researching and developing my research ideas. Reviewing literature on AI, EEG data, personalization, the Metaverse, the HCI wave, and self-discrepancy theory was essential to gain a multifaceted understanding of the factors influencing my research direction. Simultaneously, in the first phase of my first year, I researched cozy games online [17].

From an academic perspective, this study used netnography to provide thematic insights into player expectations and immersive experiences from both player and researcher viewpoints (Figure 1). From an applied perspective, research on cozy games helped me understand the essential elements of user experience design. It formed a crucial foundation for my second study – the co-design workshop.

In the second phase of my first year, I applied to the Nottingham School of Art and Design for ethics approval to conduct direct research with participants. After approval, I conducted a workshop in the form of information sharing with various groups of participants in the early stages of creative practice on a metaverse generated by AI and EEG data. I used interview questions and idea-sketching activities during this phase. From an academic perspective, I used thematic analysis to examine participants' perspectives in details. I coded the results from players' sketches to inform a research and development framework for user experience in interactive virtual environments, using physiological data as input. From an

application perspective, this research reveals several important characteristics of participants' expectations and fears regarding the interactive Metaverse experience created by GenAI, with EEG as input data in the future (Figure 2).

In the third phase of my first year, I developed a simultaneous interactive screen and VR experience, which followed our *closed-loop immersive model* - a model that securely takes the user's brain electrical data as continuous input for real-time experiences to test its feasibility for the experimental design in the following study (Figure 3). Study 3 involved in-depth interviews with seasoned professionals in the creative and arts fields to understand their perspectives on the application of new technologies, specifically GenAI, in creativity and its relationship to critical thinking in artistic creation. Theoretically, this research contributes to experts' understanding of the unique human element and the experience that AI or any other technology cannot replace, expanding perspectives on user-computer interaction. In practical terms, the findings from this study on how humans use critical thinking and bring real-world experience into artistic inspiration form a crucial foundation for my user experience design in subsequent studies.

In the fourth phase of my first year, I conducted a pilot study to test an experience I created using StreamDiffusion [9] to generate real-time visual experiences, with EEG as input data. I designed the experience with a user-centric approach, using personalized prompts from the user; connecting the user's mental experience through a two participants interactive experience and simultaneous

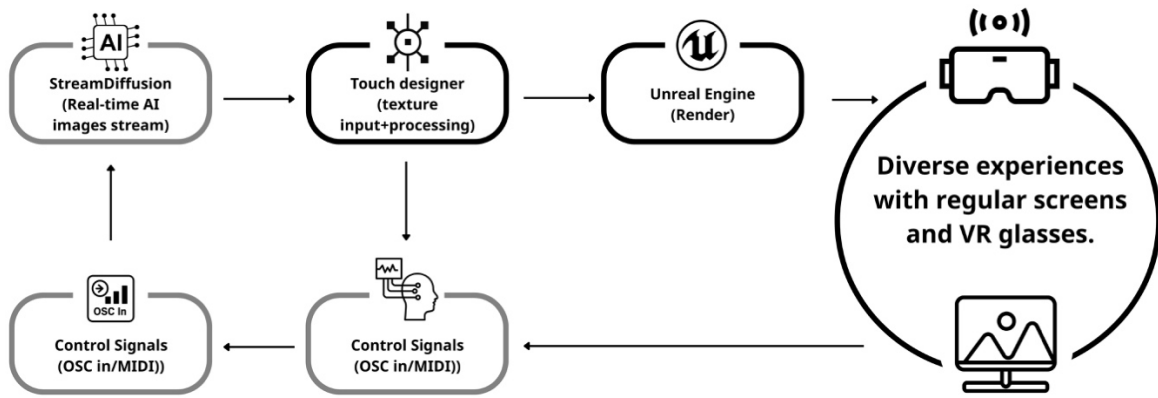


Figure 3: Closed-loop immersive model

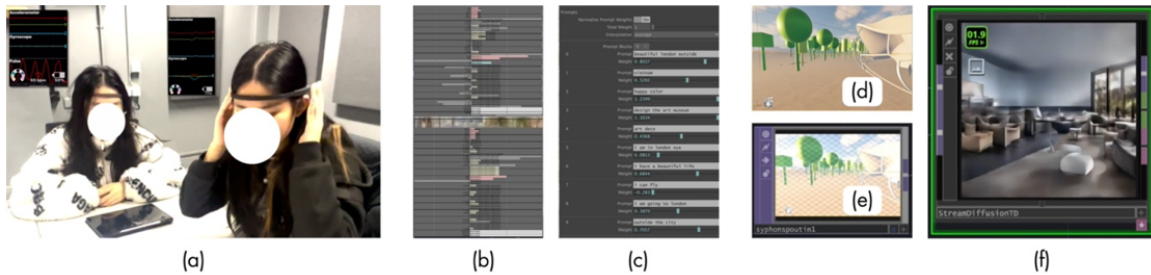


Figure 4: (a) shows participants wearing EEG headbands and data displayed on the Mind Monitor app screen. (b) shows EEG data extracted from Mind Monitor into OSCin in TouchDesigner. (c) shows OSCin data pasted as a reference for the weight of prompts provided by participants, which serves as input data. (d) shows the environment in Unreal Engine. (e) shows the Unreal Engine environment being used as one of the input sources for the GenAI-generated experience. (f) shows the continuously generated GenAI imagery

EEG data synthesis; and ensuring the experience was responsive and real-time with EEG as input data (Figure 4). Theoretically, the research results, presented in both qualitative and quantitative forms, showed a consistent, integrated relationship that strengthened the current hypothesis that EEG can be used as input data to generate real-time user experiences. In practical terms, this research makes a significant contribution to the application of participant experience deployment, real-time data collection, and data analysis, which will serve as the foundation for the main experiment in my doctoral dissertation.

The main research components of my thesis include a literature review and five studies. To date, I have completed the literature review and four studies. Across five studies in this thesis, I have presented findings demonstrating the validity and feasibility of EEG and GenAI in transforming UX from its original form into neuro-UX through brain-computer interaction. The development, grounded in these studies, has established the connection and provided an argument for future research using closed-loop immersive models. By CHI2026, I will present the completed sections and add a section on the results of Study 5 – Main Experiment to provide a comprehensive overview of the research outcome. Study 1 – Poster paper: From Cosy Games to Metaverse: Deriving Positive Interaction Qualities Through Netnography, accepted and presented at

ARPPID 2025, forthcoming publication in Academic Research and Professional Practice in Interaction Design.

5 Next steps

In my next year of doctoral studies, I focused primarily on user experience design. I aimed to design an experience that combines virtual and real environments, creating a hybrid space to serve the following purposes: (1) The ideal self is a personal element of each user, serving the real-world element; (2) the experience is generated in real time by GenAI and changes based on input EEG and eye-tracking data from the virtual environment. Theoretically, this research contributes to HCI an AI-assisted experience that generates real-time feedback based on EEG data, with eye-tracking as input data to prove the hypothesis that "If the user-AI experience is designed correctly, it can reduce the 'brain rot' caused by social media or short content, and it can increase user focus and positive mental state." In practice, I need to address in the code: (1) combining eye-tracking to identify the area the user is focusing on; (2) training the AI so that the system understands real-time EEG data. This study aims to provide robust findings demonstrating how differences in participants' ideal selves affect their experiences in the Metaverse. The EEG data's diversity reflects user experiences

and aims to observe increases in participants' immersion and focus during the GenAI-generated experience.

I will develop the narrative for my PhD thesis in my third year. My future research aims to bridge the gap between technological differentiation and access to EEG data as personalized input for all interactions and experiences, while enabling a more diverse and seamless interactive experience that supports simultaneous connections for a greater number of users.

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